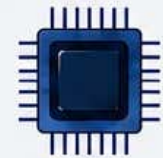
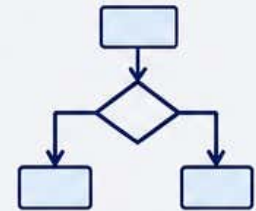




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Syllabus



Major Course: Digital System and Microprocessor

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	112223	Digital System and Microprocessor	2	30	2 Hrs.	30

Course Objectives:	1. To introduce students to number systems, binary arithmetic, and coding techniques used in digital electronics and microprocessor systems. 2. To develop understanding of logic gates, Boolean algebra, and combinational logic circuits for digital system design. 3. To enable students to simplify Boolean expressions and solve logic design problems using Boolean laws and K-Maps. 4. To familiarize students with the basic architecture, organization, and functioning of the 8085 Microprocessor and microcomputer system.						
Course Outcomes:	Students will be able to – 1. Understand different number systems. 2. Analyze and implement digital logic circuits using basic and universal logic gates, adders, and combinational circuits. 3. Apply Boolean algebra and K-Map techniques to simplify Boolean expressions and design optimized digital circuits. 4. Understand the architecture and Pin configuration of the 8085						
Unit System	Contents				Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies
Unit I	Binary Arithmetic: Binary, Octal & Hexadecimal number system and their inter conversion, Binary arithmetic: Addition and Subtraction using 1's & 2's compliments, Multiplication & Division. Binary codes: 8421 BCD, Excess-3 & Gray code.				8 Hrs.	8 Marks	Use any pedagogical technique suitable for the topic
Unit II	Logic Gates: Definition and truth tables of NOT, OR, AND, NAND, NOR gates. EX-NOR & EX-OR gates, Universal gates, Designing of Basic gates using NAND and NOR gates, Half adder, Full adder, 4-bit binary full adder.				7 Hrs.	7 Marks	
Unit III	Boolean Algebra: Boolean laws, De-morgan's theorem, Simplification of Boolean equations using				8 Hrs.	8 Marks	

	Boolean algebra, Fundamental Products & Sum terms, K-maps: up to 2,3,4 variables.			
Unit IV	Microprocessor: Block diagram of microcomputer, Evaluation of microprocessor, Architecture of 8085 Microprocessor, 8085 pin Microprocessor.	7 Hrs.	7 Marks	
References:	Course Material/Learning Resources Text books: 1. Digital and analog technique by Navneet, Kale and Gokhale (Kitab mahal prakashan) 2. Introduction to digital electronics by Mohinder Singh 3. Digital principle and application by Malvino and Leach 4. Microprocessor Architecture, Programming and Applications with 8085 by Ramesh S. Gaonkar Reference Books: 1. Pulse, digital and switching waveforms by Millman and Taub 2. Principle of electronics by V. K. Mehta 3. Basics of electronics by B. L. Theraja 4. The 8085 Microprocessor: Architecture, Programming and Interfacing by K. Udaya Kumar, B. S. Umashankar Weblink to Equivalent MOOC on SWAYAM if relevant: https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_ee125 https://onlinecourses.swayam2.ac.in/e-learning/preview/cec21_cs16 Weblink to Equivalent Virtual Lab if relevant: https://www.vlab.co.in/broad-area-electronics-and-communications Any pertinent media (recorded lectures, YouTube, etc.) if relevant: https://www.youtube.com/watch?v=jte2S2iWRMI https://www.youtube.com/watch?v=1oHz2svCOC8			

Assessment Rubric: Internal for Theory Course (Major)

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI THEORY EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Theory	Course Code: 112223	Course Title: Digital System and Microprocessor	Max Marks: 20
Sr. No.	Assessment Criteria		Marks
1	Any Two of the Activity (weekly or monthly assignment/ Unit Test/Class Test) (CAT-1)		10
2	Any One of the Activity includes Open Book Test, MCQ Tests, / Theory related Class activity/Task etc.) etc. (CAT-2)		5
3	Participation in any one Activity includes Quiz, Seminar, Group Discussion, Industry visit, Case Study, Mini project, innovative work, etc. based on Theory Course. Involvement in subject related activities such as Conference, Workshop, Coding Competitions etc. (CAT-3)		5